1. **Under pressure: Interference** (from G & G Ch. 8): In their study of spillover effects, Sinclair, McConnell, and Green[1] sent mailings to randomly selected households in Chicago encouraging them to vote in an upcoming special election on April 7th 2009 (Figure 1). The mailings used a form of social pressure, disclosing whether the targeted individual had voted in previous elections. Because this type of mail had proven to increase turnout by approximately 4-5 percentage points in previous experiments, the authors used it to study whether the treatment effects are transmitted across households.

   Employing a multilevel design, they randomly assigned all, half, or none of the members of each nine-digit zip code to receive mail. For purposes of this example, we focus only on households with one registered voter. The outcome variable is voter turnout as measured by the registrar of voters. The results are as follows. Among registered voters in untreated zip codes, 1,201 of 6,217 cast ballots. Among untreated voters in zip codes where half of the households received mail, 526 of 3,316 registered voters cast ballots. Among treated voters in zip codes where half of the households received mail, 620 of 2,949 voted. Finally, among treated voters in zip codes where every household received mail, turnout was 1,316 of 6,377.

   a. Identify the response variable, the units (measurement and experimental), and the experiental factor(s) and their levels.

   b. Using potential outcomes, define the direct treatment effects (parameters) of receiving mail addressed to subject i.

   c. Define the spillover treatment effects (also parameters) of being in a zip code where varying fractions of households are treated.

   d. Propose an estimator for estimating the firsthand (direct) and secondhand (spillover) treatment effects. Show that the estimator is unbiased, explaining the assumptions required to reach this conclusion.

   e. Based on these data, what are your estimates of the magnitude of the mailing's direct and spillover effects? What did we learn here scientifically?

[1] Sinclair, McConnell, and Green (2012). *"Detecting Spillover Effects: Design and Analysis of Multilevel Experiments"* in *American Journal of Political Science.*

FIGURE 1 Sample Mailer

Dear Richard L Jensen:

DO YOUR CIVIC DUTY AND VOTE ON APRIL 7!

Why do so many people fail to vote? We've been talking about this problem for years, but it only seems to get worse -- especially when elections are held in the spring.

This year we're taking a different approach. We're reminding people that who votes is a matter of public record. The chart shows your name from the list of registered voters and whether you voted in the last two spring elections. The chart also contains an empty space that we will fill in based on whether you vote in the April 7th election.

DO YOUR CIVIC DUTY AND VOTE ON APRIL 7!

VOTER NAME	Spring 2006	Spring 2008	April 7
RICHARD L JENSEN	Didn't Vote	Didn't Vote	_______

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Figure 1: Example of postcard mailed to registered voters.

More questions coming soon!